Generics Assignment

Food.java:

```
public class Food {
  private String name;
  private double weight; // pounds

  public Food(String name, double weight) {
    this.name = name;
    this.weight = weight;
  }

  public String getName() {
    return name;
  }

  public double getWeight() {
    return weight;
  }

  @Override
  public String toString() {
    return name + " (" + weight + " lb)";
  }
}
```

Bread.java:

```
public class Bread extends Food {
  public Bread(String name, double weight) {
    super(name, weight);
  }
}

class WhiteBread extends Bread {
  public WhiteBread(double weight) {
    super("White Bread", weight);
  }
}
```

```
class Brioche extends Bread {
  public Brioche(double weight) {
    super("Brioche", weight);
  }
}

class Baguette extends Bread {
  public Baguette(double weight) {
    super("Baguette", weight);
  }
}
```

Fruit.java:

```
public class Fruit extends Food {
  public Fruit(String name, double weight) {
    super(name, weight);
  }
}

class Banana extends Fruit {
  public Banana(double weight) {
    super("Banana", weight);
  }
}

class Mandarin extends Fruit {
  public Mandarin(double weight) {
    super("Mandarin", weight);
  }
}

class Avocado extends Fruit {
  public Avocado(double weight) {
    super("Avocado", weight);
  }
}
```

Vegetable.java:

```
public class Vegetable extends Food {
  public Vegetable(String name, double weight) {
    super(name, weight);
  }
}

class Onion extends Vegetable {
  public Onion(double weight) {
    super("Onion", weight);
  }
}

class Cilantro extends Vegetable {
  public Cilantro(double weight) {
    super("Cilantro", weight);
  }
}

class Tomato extends Vegetable {
  public Tomato(double weight) {
    super("Tomato", weight);
  }
}
```

Bucket.java:

```
import java.util.ArrayList;

public class Bucket<T> {
  private ArrayList<T> list;

  public Bucket(ArrayList<T> list) {
    this.list = list;
  }

  public void add(T item) {
    list.add(item);
```

```
  }

  public ArrayList<T> getList() {
    return list;
  }

  @Override
  public String toString() {
    return "Bucket{" +
        "list=" + list +
        '}';
  }
}
```

Main.java:

```
import java.util.ArrayList;

public class Main {
  public static void main(String[] args) {

    Fruit banana = new Banana(0.25);
    Fruit mandarin = new Mandarin(0.20);
    Fruit avocado = new Avocado(0.45);

    Vegetable onion = new Onion(0.35);
    Vegetable cilantro = new Cilantro(0.10);
    Vegetable tomato = new Tomato(0.40);

    Bread white = new WhiteBread(1.10);
    Bread brioche = new Brioche(1.00);
    Bread baguette = new Baguette(0.55);

    ArrayList<Food> mixedList = new ArrayList<>();
    Bucket<Food> bucket = new Bucket<>(mixedList);

    bucket.add(banana);
    bucket.add(mandarin);
    bucket.add(avocado);
```

```
        bucket.add(onion);
        bucket.add(cilantro);
        bucket.add(tomato);

        bucket.add(white);
        bucket.add(brioche);
        bucket.add(baguette);

        System.out.println("Items in the bucket:");
        for (Food item : bucket.getList()) {
            System.out.println(item);
        }

    }
}
```

Output:

```
"C:\Program Files\Java\jdk-21\bin\java.exe"
Items in the bucket:
Banana (0.25 lb)
Mandarin (0.2 lb)
Avocado (0.45 lb)
Onion (0.35 lb)
Cilantro (0.1 lb)
Tomato (0.4 lb)
White Bread (1.1 lb)
Brioche (1.0 lb)
Baguette (0.55 lb)

Process finished with exit code 0
```